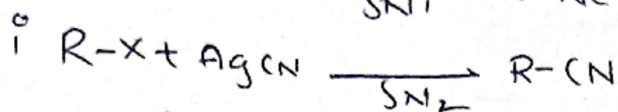
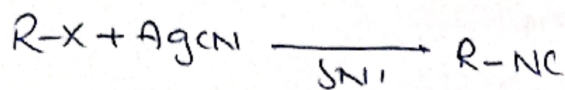
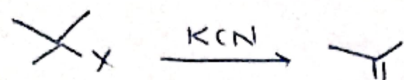
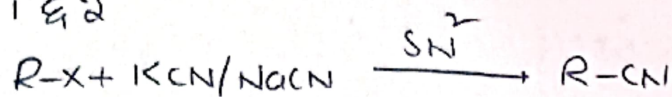
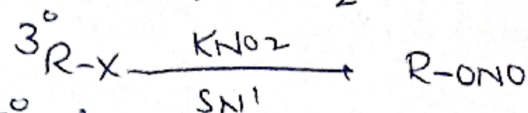
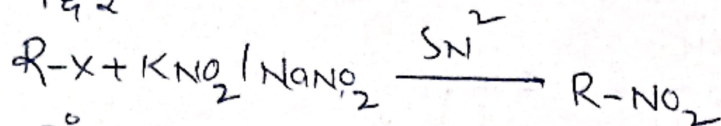


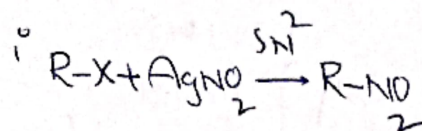
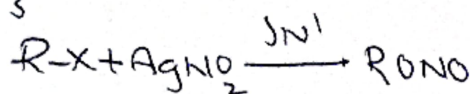
1° & 2°



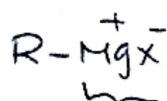
1° & 2°



2° & 3°



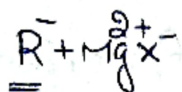
*** GRIGNARD REAGENT $RMgX$



Organic Metallic Reagent

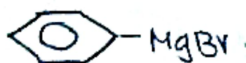
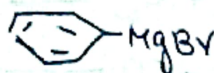
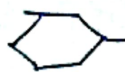
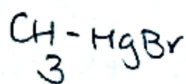
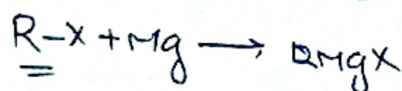
~ Ionic Bond

Covalent

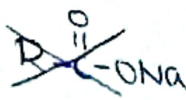


~ Alkyl Part Always acts like NCI^-

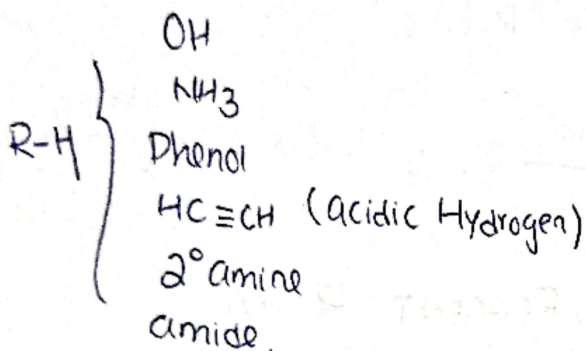
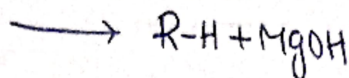
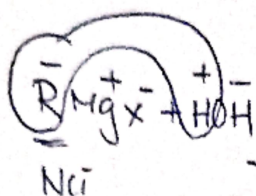
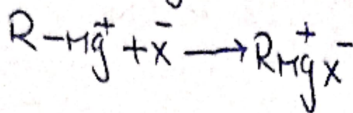
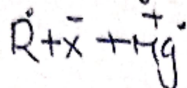
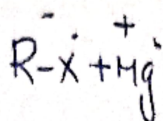
- 1) EPS (Electrophilic Sub Rxn)
- 2) Sub
- 3) Elimination



- Metal Carbon tho attach undalee.



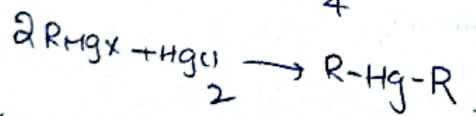
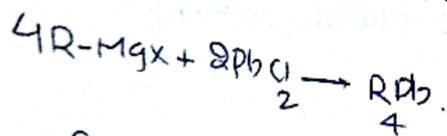
} NOT
Organic Metallic Reagent



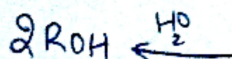
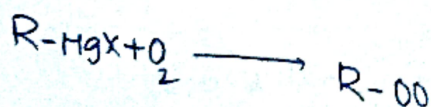
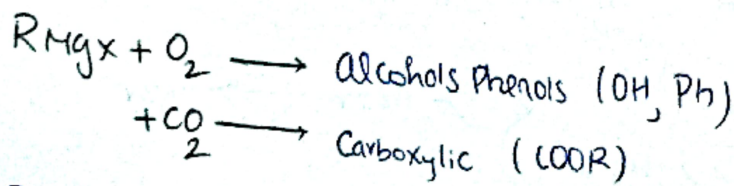
- Electrophilic Subst Rxn.

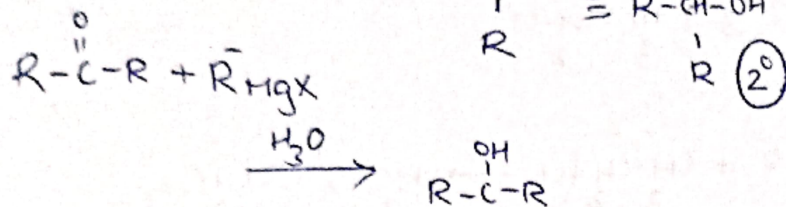
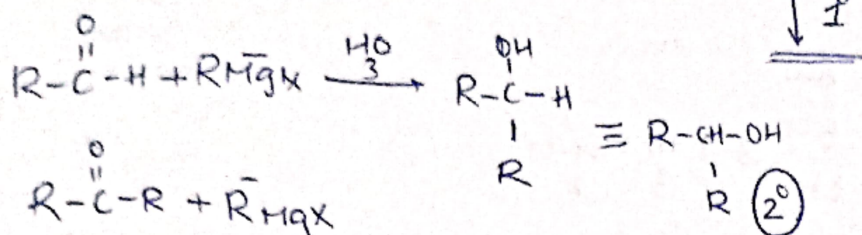
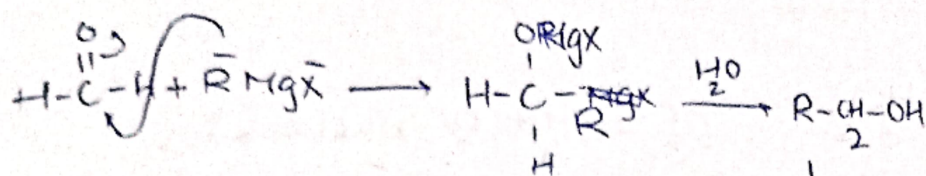
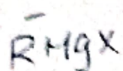
TEL (Tetra ethyl lead)
TML (Tetra methyl lead) } Antiknocking Agent

- led is carcinogenic

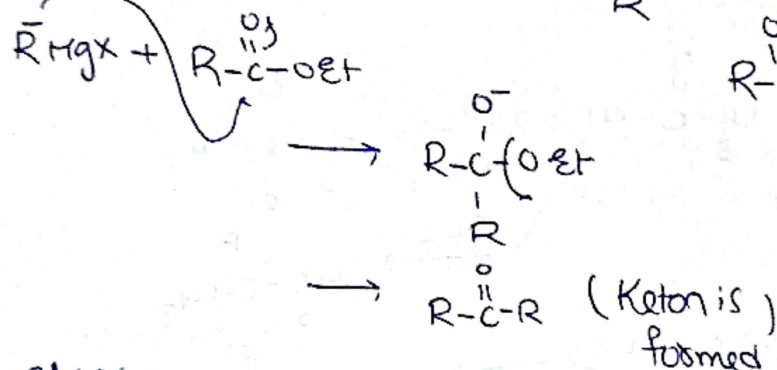
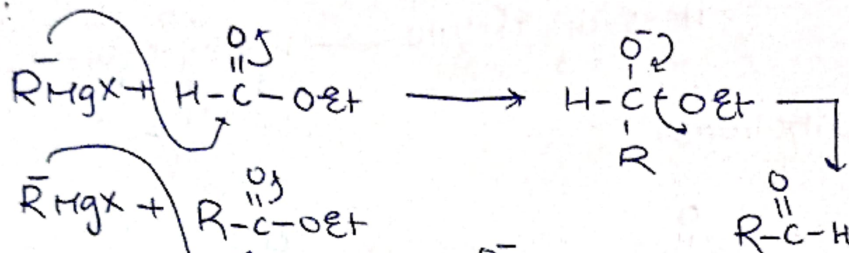


#> Trick

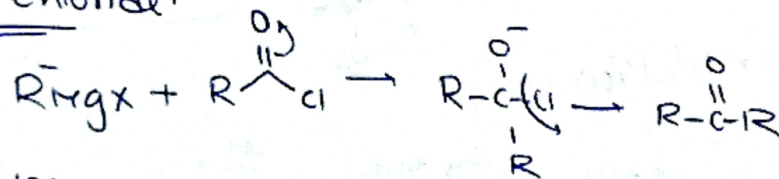




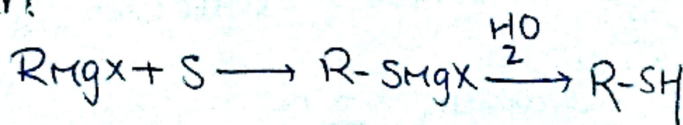
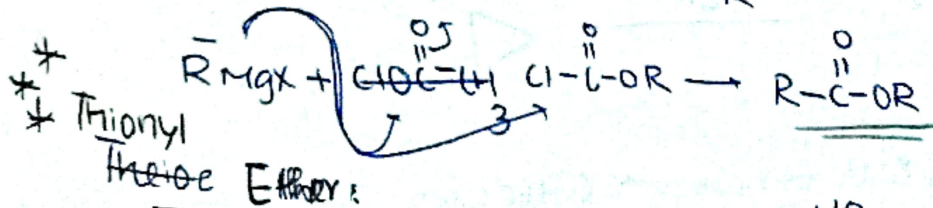
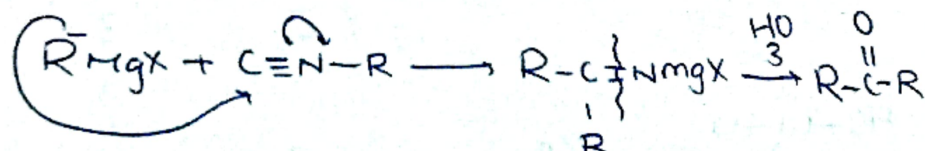
Ester:



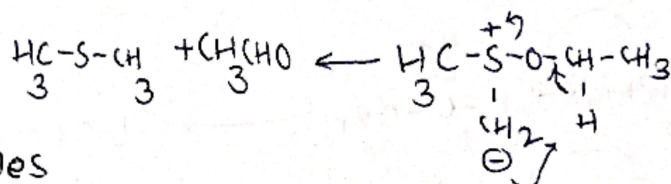
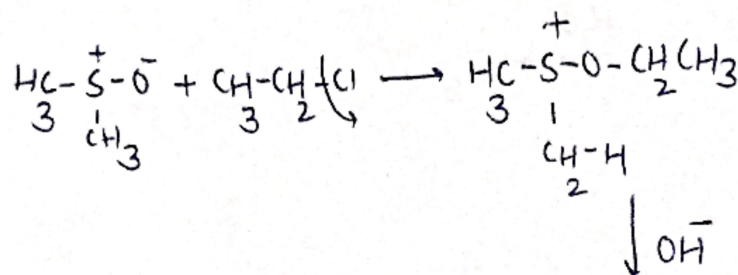
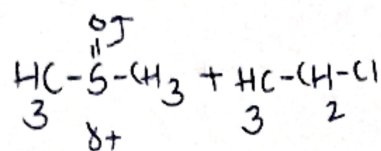
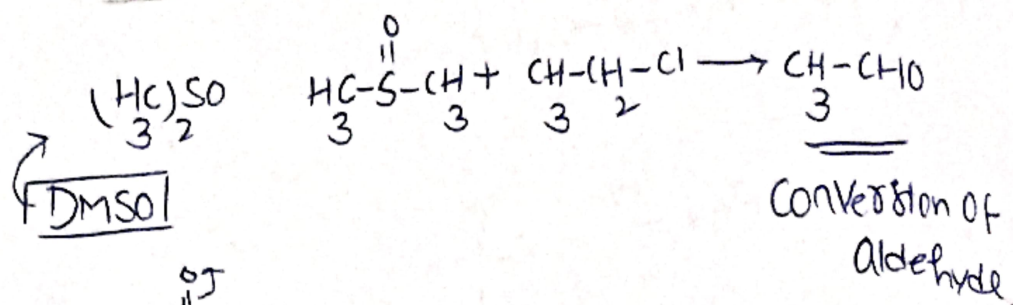
Acid chloride:



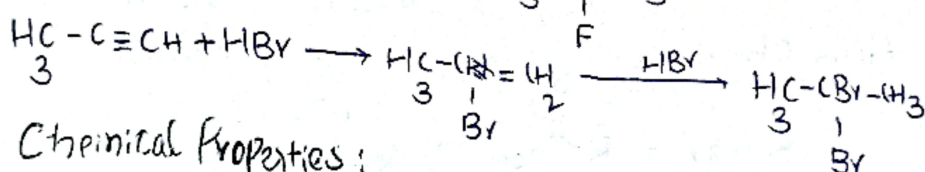
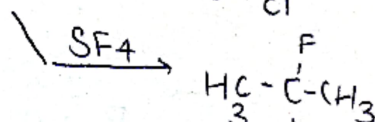
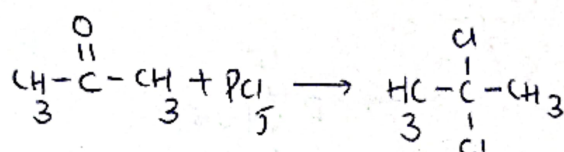
Cynide:



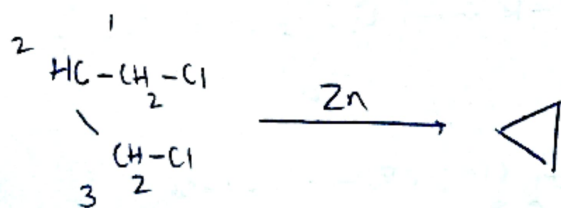
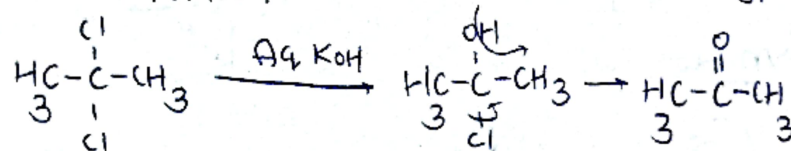
Oxidation of R-X:



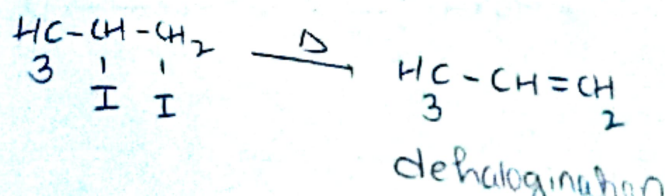
Dihalides



Chemical Properties:



1,3 & 1,6 dihalo $\rightarrow$ Cyclic compounds



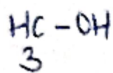
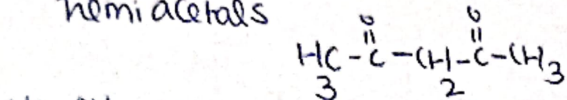
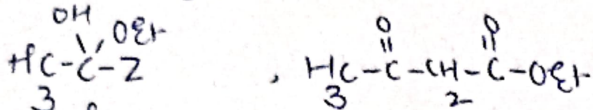
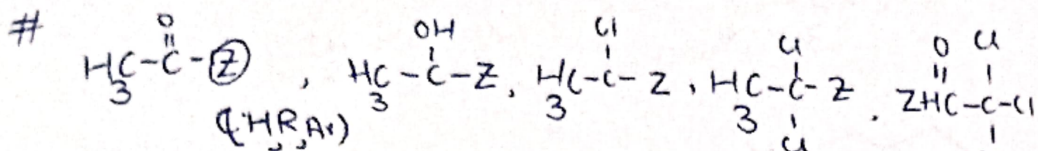
Haloform Reaction.

to Oxyanion containing Halogen

* Formation of Haloform from organic compound

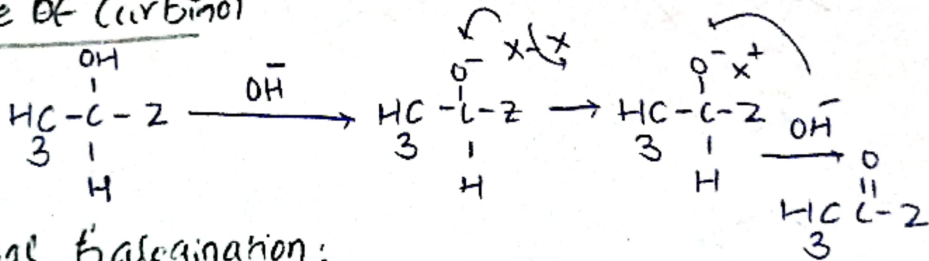
When it is exposed to halophalites is known as

Haloborn reaction

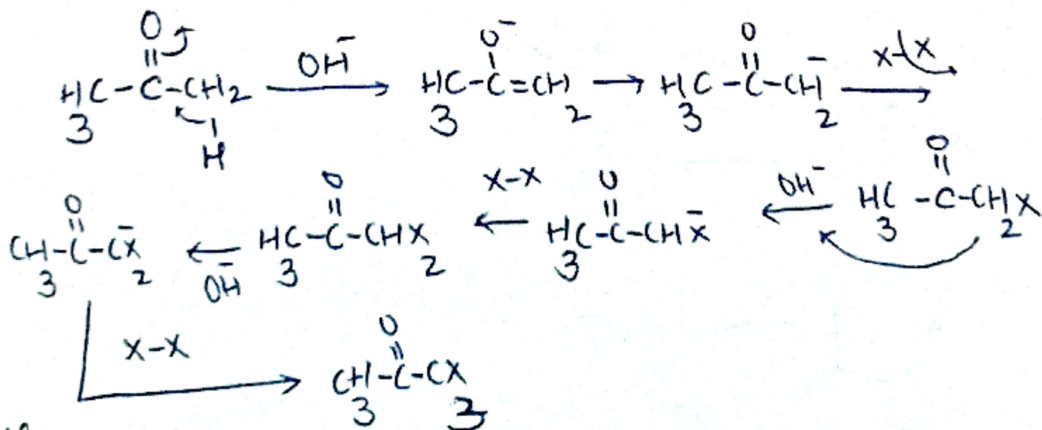


* Acids & Acid derivatives will NOT give Haloform.

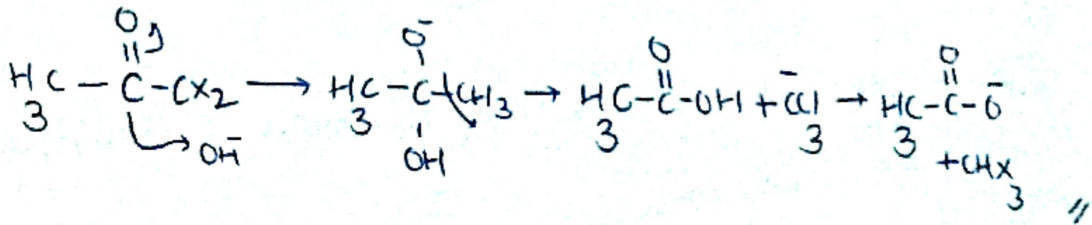
① Oxidative of Curbinol

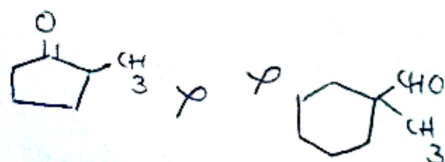
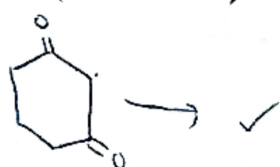
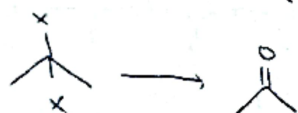
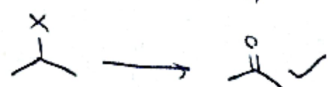
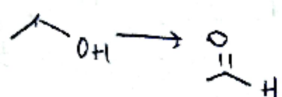
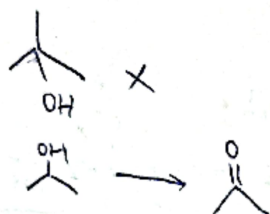
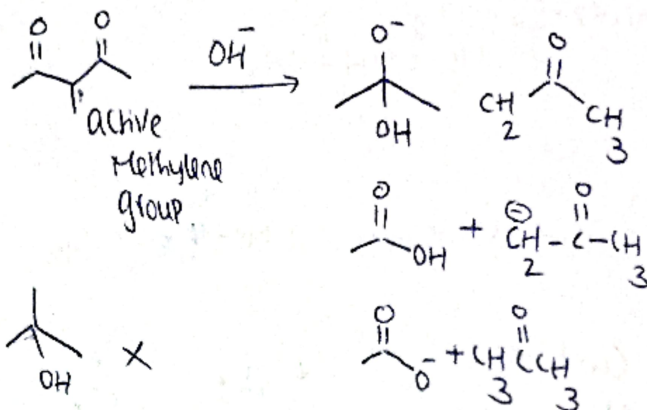
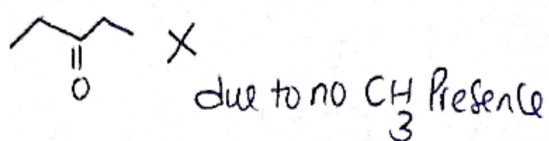
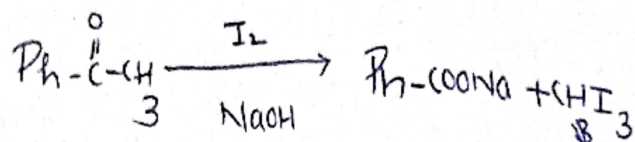
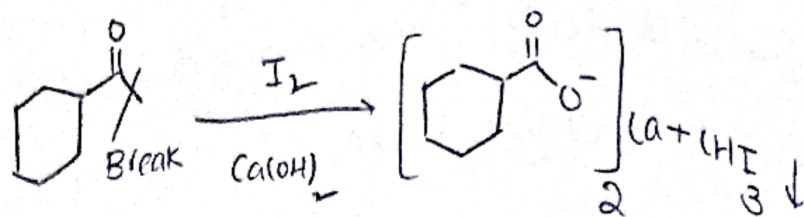
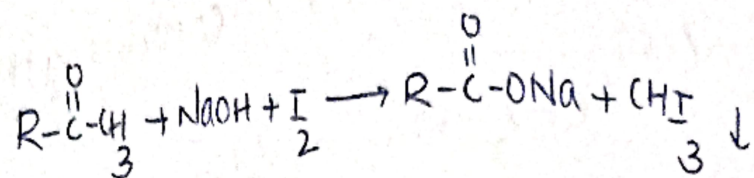


② Sequential Halogenation:



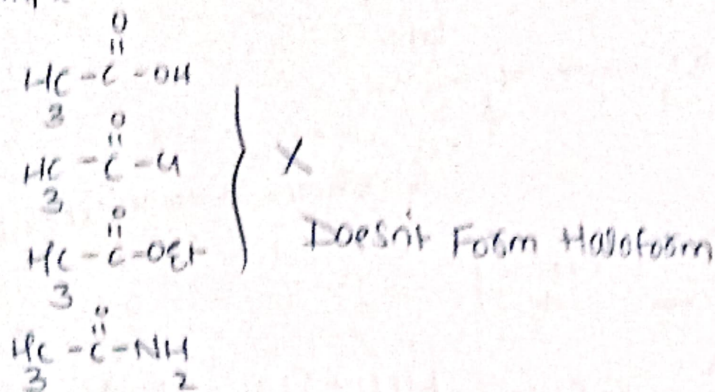
⑤ Hydrolysis of trihalo carbonyl.





⇒ acidic hydrogen Substitution
 = Hydrogenation Urea compound ke Haloform
 jaragadu...

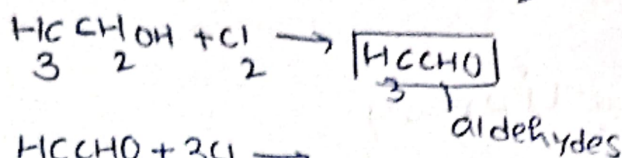
Example:



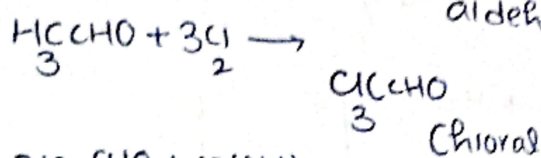
Preparation of Chloroform:

Ethanol/Acetone + Bleaching Powder

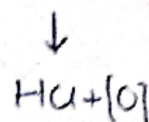
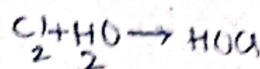
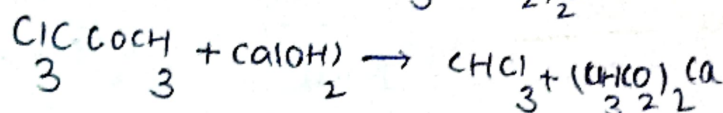
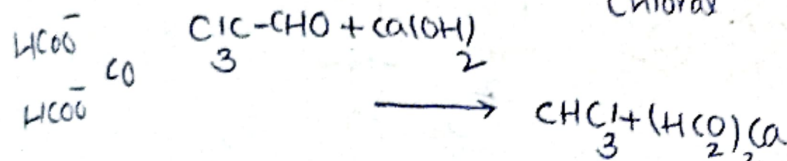
or Cl_2 in alkalis $\rightarrow$ Chloroform



aldehydes

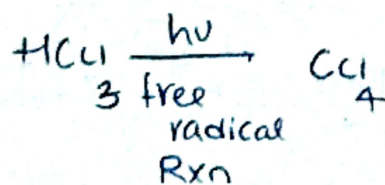
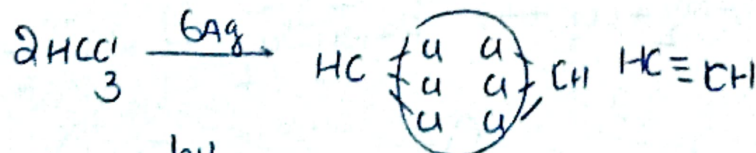
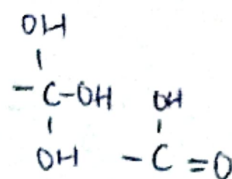
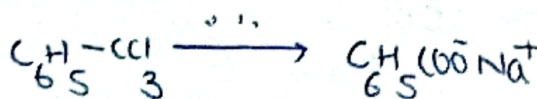
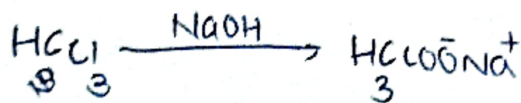


Chloral

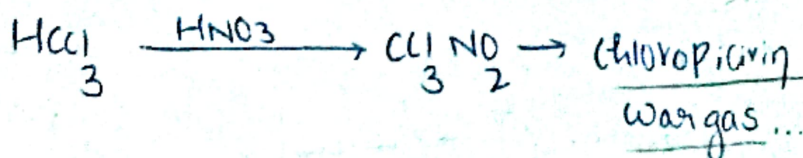


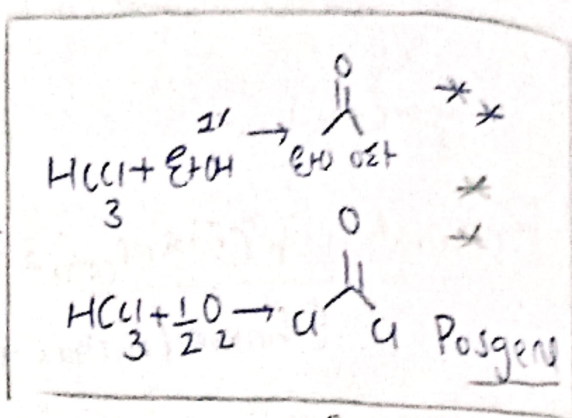
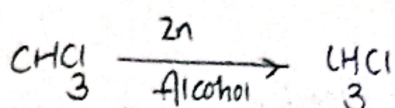
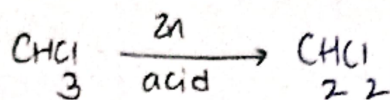
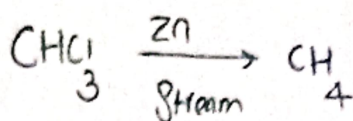
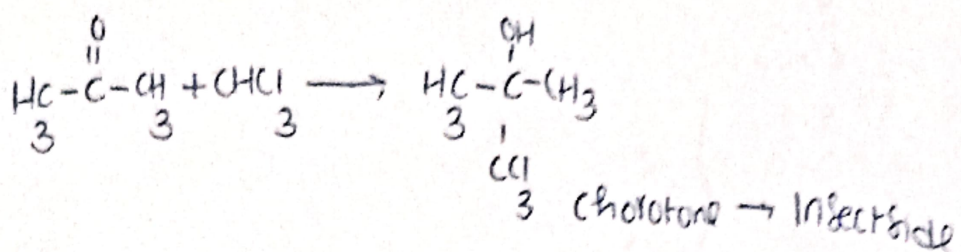
Nascent oxygen

Chemical reactions:



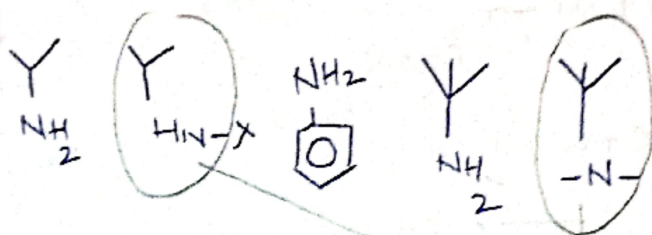
free radical Rxn.





Carbyl Amine (or) Isocynide or Test for 1° amines:

* RNH_2 ArNH_2 (1° Amines)



1° amine $\xrightarrow[\text{alcohol}]{\text{CHCl}_3}$ Isocynide Can't form

